

Deep Learning Approaches for Environmental Sound Classification (ESC)

Prashant Sah Karan Shah Jenny Li Rahul Sati
Georgia Institute of Technology
225 North Ave NW, Atlanta, GA 30332

psah8@gatech.edu, kshah498@gatech.edu, jli3495@gatech.edu, rsati3@gatech.edu

Abstract

This study investigates the factors influencing deep learning performance in Environmental Sound Classification (ESC) using the ESC-50 dataset. An ablation study was conducted to evaluate three distinct input representations: raw waveforms, Mel-spectrograms, and Mel-Frequency Cepstral Coefficients (MFCCs). The results demonstrate that Mel-spectrograms provide the most informative representation, achieving a validation accuracy of 72.8%, significantly outperforming MFCCs (67.8%) and raw waveforms (54.8%). Multiple neural architectures were then tested, including a baseline CNN and pre-trained image models like ResNet-18 and EfficientNet-B0. While both pre-trained models learned strong features, EfficientNet-B0 exhibited superior generalization over ResNet-18, which showed a higher tendency to overfit. Spectral augmentation was then integrated with the EfficientNet-B0 model, achieving a validation accuracy of 81%, rivaling the established human baseline accuracy. These findings suggest that leveraging transfer learning from image-recognition models paired with spectral-based inputs and strategic augmentation offers a computationally efficient and effective path for real-world environmental sound recognition.

1. Introduction/Background/Motivation

This study aims to develop a system capable of recognizing everyday environmental sounds such as dog barks, rainfall, door closures, and distant sirens. Although these sounds are common in real-world settings, they can present a challenge for machine learning methods to classify accurately due to variation in loudness, duration, and background noise. The primary objective of this work is to determine the factors that most influence the performance of the model. Rather than evaluating a single model, this study focused on analyzing and exploring why certain choices matter. Specifically, three questions were considered: a). Will the model perform better when trained on raw audio signals or on image-based representations such as spectrograms?

b). Do larger pre-trained models, originally designed for image recognition tasks, outperform a smaller manually-trained CNN model? c). Which design decisions, including input representation and training strategy contribute most significantly to classification accuracy? Overall, the project breaks down the problem of environmental sound recognition into smaller, measurable questions in order to better understand what works, what does not, and why. Our objective was not solely to maximize accuracy, but also to identify which approaches improve performance and which ones offer limited value.

Environmental sound recognition is currently performed using a range of machine learning and deep learning techniques trained on labeled audio datasets. The most common approach is to transform the audio signals into structured representations such as Mel spectrograms, log-Mel spectrograms, or MFCCs, which capture time-frequency characteristics. These representations are then processed using architectures such as CNNs, RNNs, LSTMs, Transformers, or hybrid CNN-Transformer models. Transfer learning is a wide practice currently, where models are pre-trained on large image or audio datasets and then fine-tuned for classification. Additionally, self-supervised learning methods are also becoming increasingly common, where large volumes of unlabeled data are learned by the models before supervised training. Although current methods are able to achieve strong results, several limitations remain. First of all, many such systems perform well under controlled conditions but experience reduced accuracy and generalization in real-world environments, where background noise, overlapping sound events, reverberation, and low-quality is common in the recordings. Secondly, high-performing models also often require high computational resources, which limits their deployment on mobile or embedded devices. Thirdly, many approaches depend on large labeled datasets, which are costly and time-consuming to create.

Environmental sound classification has a strong real-world impact across multiple domains. Smart home devices like Amazon Echo or Google Nest respond to voice commands but lack awareness of surrounding sounds; detect-

ing events like smoke alarms, baby cries, or breaking glass could enable automatic safety responses. For the 466 million people in the world with disabling hearing loss, sound recognition on phones or hearing aids could restore critical environmental awareness.

Beyond homes, cities use acoustic monitoring to track noise pollution and detect events like gunshots or crashes, but current systems rely on costly hardware and manual analysis. A lightweight, accurate classifier using standard audio inputs could make this scalable and accessible. In industrial settings, monitoring machine sounds can help detect failures early, reducing downtime and safety risks.

The key contribution is efficiency: using ImageNet-pretrained CNNs on Mel-spectrograms achieves around 80% accuracy without audio-specific pretraining or massive datasets. This lowers barriers to deployment, making sound classification feasible on edge devices, embedded systems, and in low-resource environments.

The data used for this study was the ESC-50: Dataset for Environmental Sound Classification. The data set consists of 2,000 labeled audio recordings structured into 50 semantic classes with 40 examples per class. The 50 semantic classes are organized into 5 categories: animals, natural soundscapes and water sounds, human non-speech sounds, interior/domestic sounds, and exterior/urban noises. The audio clips were extracted from public field recordings provided by the Freesound.org project with the goal of providing a standardized, high-quality, publicly available dataset for non-speech and non-music audio classification. Each audio clip is 5 seconds long with a 44.1 kHz sampling rate. The raw data is a 1D waveform, but for CNN based models, it is recommended to transform the data into a 2D Mel-spectrogram. The dataset is already split into 5 folds to facilitate cross-validation. There are dozens of publications with accuracy ranging from 32% to 98%, with human accuracy of 81% that can be used as a benchmark for this study.

2. Approach

To solve the problem of recognizing everyday environmental sounds, we designed our approach around the idea that understanding how a model learns is just as important as the final accuracy it achieves. Instead of relying on a single model or representation, we built a structured pipeline that allowed us to explore the problem from multiple angles. Our first step was to decide how to represent the audio itself. Raw audio is simply a long sequence of numbers, which can be difficult for a model to interpret directly, so we experimented with three different forms of the same sound: the raw waveform, a Mel-spectrogram, and MFCC features. The Mel-spectrogram converts the sound into a two-dimensional image-like pattern showing how energy changes across frequencies over time, while MFCCs compress this information into a more compact form often

used in speech processing. By comparing these representations under identical training conditions, we could determine which form of the sound carried the most useful information for classification.

Once we had these representations, we trained several types of models to understand how model architecture influences performance. We began with a small convolutional neural network that we built ourselves, which served as a baseline and helped us understand what performance looks like without any external knowledge. We then trained two large pretrained image models: ResNet-18 and EfficientNet-B0 that were originally trained on millions of natural images. Because Mel-spectrograms resemble images, we believed these pretrained models might transfer their ability to detect visual patterns to the audio domain. To complement these, we also built a one-dimensional convolutional network specifically designed for raw waveforms, allowing us to compare image-based learning with direct audio-based learning. Training all of these models under the same conditions gave us a controlled way to evaluate how much each architectural choice contributed to performance.

To make the models more robust and better prepared for real world variability, we incorporated several forms of data augmentation. Environmental sounds can shift in time, change in pitch, or contain missing or distorted regions, so we introduced augmentations that mimic these variations. Time shifting moves the audio forward or backward, pitch shifting raises or lowers the sound, and SpecAugment masks random time or frequency regions in the spectrogram. These augmentations force the model to focus on the essential characteristics of each sound rather than memorizing exact patterns. When we applied these augmentations to EfficientNet-B0, the model's validation accuracy improved to 81%, demonstrating that the model became more general and less sensitive to small changes in the input.

Throughout the project, we kept the training setup consistent across experiments so that comparisons would be fair and meaningful. This allowed us to answer deeper questions about the problem, such as whether raw audio or spectrograms are more informative, whether large pretrained models truly outperform smaller custom ones, and whether augmentations actually improve generalization. We believed this approach would be successful because it combined the strengths of both audio-specific techniques and powerful pretrained image models, while also giving us insight into why certain choices work better than others. What makes our approach distinctive is that we did not rely on a single model or representation; instead, we built a full ablation study that systematically compared multiple input types, architectures, and training strategies. This gave us a clearer understanding of the underlying factors that drive performance in environmental sound classification, rather

than simply reporting a final accuracy number.

During this project, we expected several difficulties, especially because environmental sounds vary widely in length, loudness, and background noise, and we were unsure which representation: raw audio, spectrograms, or MFCCs would actually give the models enough structure to learn from. We also anticipated that training deep models on a relatively small dataset like ESC-50 might lead to overfitting, and that adapting large pretrained image models to work with single-channel spectrograms could introduce technical complications. In practice, many of these challenges did appear. Our first attempts at data augmentation did not work as intended, the raw waveform model struggled more than expected, and the earliest training runs of the baseline CNN produced unstable accuracy until we refined the pre-processing pipeline. The very first thing we tried certainly did not work perfectly, but each failure helped us better understand the problem and guided us toward more effective choices, such as using Mel-spectrograms, fine-tuning pretrained models, and adding time, pitch, and spectrogram based augmentations to improve generalization.

Here is the link for our entire codebase: <https://colab.research.google.com/drive/1u8jWbeu0q-W3CI46Utv1E0t-TPOGeHP6?usp=sharing>

3. Experiments and Results

The success of our experiments was measured through accuracy. There are dozens of publications documenting various strategies and their accuracies, and our goal was to surpass the baseline CNN accuracy of 66.0% and approach or surpass the human classification accuracy of 81.3%. The experiments were structured in 3 phases: transformation, model selection, and augmentation.

To isolate the effect of input representation on classification performance, we held every training variable constant across three experiments: 50 epochs, Adam optimizer at $lr=1e-3$, ReduceLROnPlateau scheduler (factor=0.5, patience=7), CrossEntropyLoss, batch size of 32, and an identical train/test split (Folds 2–5 training, Fold 1 test). The single variable manipulated was how raw audio was transformed before being fed to the model.

Three representations were evaluated:

- **Raw Waveform:** The 5-second audio clip at 44,100 Hz, zero-padded to a fixed length of 220,500 samples, fed directly with no spectral preprocessing.
- **Mel-Spectrogram:** Audio converted to a log-scaled Mel-spectrogram ($n_fft=1024$, $hop_length=512$, $n_mels=64$), producing a $(1 \times 64 \times 431)$ 2D feature map treated as an image.

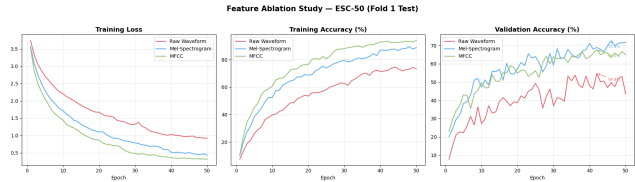


Figure 1. Ablation training curves: training loss, training accuracy, and validation accuracy across 50 epochs

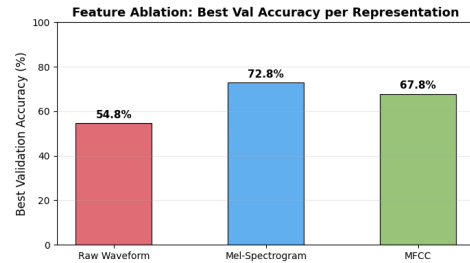


Figure 2. Best validation accuracy by input representation

- **MFCC:** Audio converted to 40 Mel-frequency cepstral coefficients using the same FFT parameters, producing a $(1 \times 40 \times 431)$ feature map.

The Mel-spectrogram performed best, reaching 72.8% validation accuracy about 5 points above MFCC and 18 above raw waveform. This was expected due to its built-in advantages. The Mel scale compresses higher frequencies in a way that matches human hearing, while dB conversion reduces dynamic range, making features more comparable. Together, these embed useful acoustic structure into the input, giving the CNN a strong inductive bias. The model was still improving at epoch 50 and showed a moderate 17-point train/val gap, suggesting reasonable generalization with room for improvement via augmentation.

MFCC achieved 67.8% but overfit heavily, with a 28.9-point gap between training and validation. Its compression step, helpful in speech tasks, removes spectral detail that is important for distinguishing environmental sounds. As a result, the model memorized training data without generalizing well.

Raw waveform performed worst at 54.8%, with a large 29.9-point gap and no clear convergence. While it still learned meaningful patterns, end-to-end audio learning requires far more data and deeper models. At this dataset size, the inductive bias from Mel preprocessing is essential.

Once we determined that Mel-spectrograms were the highest performing input representation, we expanded our model selections. In addition to the baseline CNN, we used ResNet-18 and EfficientNet-B0, which are pre-trained convolutional neural network architectures that we adapted for environmental sound classification by treating Mel-spectrograms as image-like inputs. The models were trained over 30 epochs and with a learning rate of $1e-3$.

ResNet-18 is based on the concept of residual learning, where skip connections are added between layers. These residual connections help address the vanishing gradient problem and allow the network to learn deeper feature representations more effectively. In the following experiments, ResNet-18 serves as a transfer learning model that extracts hierarchical visual features from spectrograms, such as frequency patterns and temporal structures, to help distinguish different sound classes.

EfficientNet-B0, on the other hand, was designed to achieve high accuracy along with better computational efficiency. It uses a compound scaling method that can systematically balance network depth, width and input resolution (for example, making the network deeper by adding more layers or by increasing the number of channels in a fixed, mathematically guided proportion).

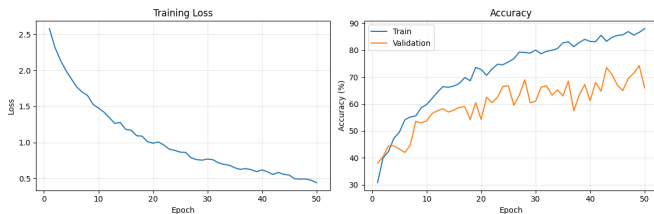


Figure 3. Training loss and accuracy ResNet-18

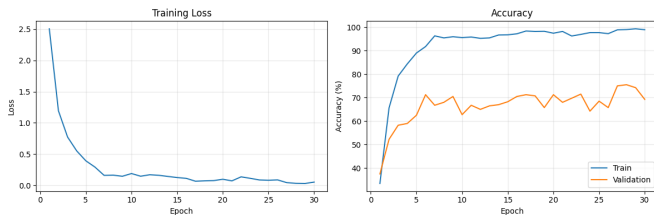


Figure 4. Training loss and accuracy ResNet-18

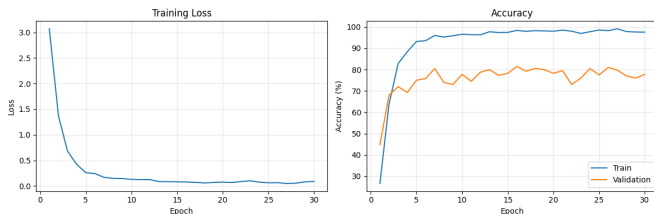


Figure 5. Training loss and accuracy EfficientNet-B0

Looking at the resulting training curves for ResNet-18 and EfficientNet-B0, both models achieve very low training loss and high training accuracy, indicating that they are able to learn strong discriminative features from the Mel-spectrogram inputs. The generalization behavior on the validation set, however, differs significantly.

ResNet-18 shows rapid improvement in training accuracy, reaching 98–99% after some epochs. However, its

validation accuracy stabilizes around the mid-to-high 60s to low 70s range with some fluctuation. This gap between training and validation performance suggests that the model overfits and struggles to generalize well on unseen audio samples. EfficientNet-B0 also achieves very high training accuracy (98–99%), but it demonstrates better validation performance overall. Its validation accuracy becomes stable closer to the upper 70s and reaches almost 80% with comparatively more stable behavior across epochs and better generalization than ResNet-18.

The difference can be attributed to the fact that ResNet-18 is relatively deep and relies heavily on residual connections, which increases the risk of overfitting on smaller datasets. In comparison, EfficientNet-B0 compound scaling prevents unnecessary model complexity as it is optimized for parameter efficiency and can extract useful high-level features without over-parameterizing.

As EfficientNet-B0 ultimately outperformed the other models, it was chosen as the starting point for further augmentation. We first introduced waveform-level transformations, specifically pitch shifting using `torchaudio.transforms PitchShift` (varying between -2 and +2 semitones) and cyclic time shifting using `torch.roll` (up to 1 second). These experiments were designed to force the model to learn invariant acoustic features—recognizing a “dog bark” or “siren” regardless of its frequency or temporal onset. However, we encountered significant technical hurdles during implementation; the computational cost exceeded the available system RAM in the Google Colab environment, causing the training process to fail before the completion of a single epoch, even after attempting to downsample the data to 16kHz and add a probability function to the implementation so that not all audio clips were augmented. We then pivoted to spectral augmentation using `torchaudio.transforms FrequencyMasking` and `TimeMasking` with a max frequency mask length of 15 bins and max time mask length of 35 bins. The primary objective of this technique was to force the model to learn invariant features by making it robust to the partial loss of spectral or temporal information. This final model achieved a validation accuracy of 81.0%, rivaling the established human baseline.

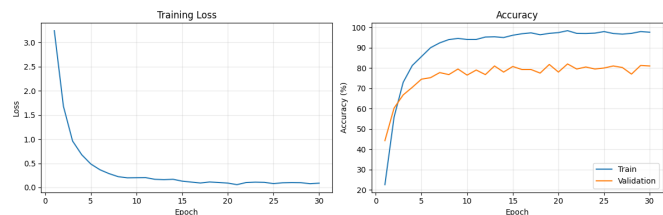


Figure 6. Training loss and Accuracy for EfficientNet-B0 with spectral augmentation

4. Conclusion & Future Work

This project set out to understand the factors that most strongly influence performance in environmental sound classification. Rather than relying on a single model or representation, we conducted a systematic study comparing raw waveforms, Mel-spectrograms, and MFCCs, along with multiple neural architectures ranging from a custom baseline CNN to large pretrained models such as ResNet-18 and EfficientNet-B0. Our experiments demonstrated that Mel-spectrograms provide the most informative representation for ESC-50, outperforming both MFCCs and raw audio by a significant margin. Among the models evaluated, EfficientNet-B0 achieved the best generalization, reaching validation accuracy near the human benchmark when combined with spectral augmentation. These results confirm that pretrained image models, when paired with spectrogram-based inputs and appropriate augmentation, offer an effective and computationally efficient solution for environmental sound classification. Overall, the study validated our initial hypothesis that both representation choice and model capacity must be carefully matched to the dataset and task, and that thoughtful augmentation can substantially improve robustness.

While our findings are promising, they also highlight several opportunities for future research. First, the limited size of ESC-50 remains a bottleneck. Larger and more diverse datasets would help reduce overfitting and improve generalization. Self-supervised learning on large unlabeled audio corpora is a particularly promising direction, as it could reduce dependence on labeled data and enable models to learn richer acoustic representations. Second, transformer based architectures such as the Audio Spectrogram Transformer (AST) have shown strong performance and merit deeper exploration, especially when combined with audio-specific pretraining. Third, real-world deployment requires models that can handle overlapping sound events, variable recording conditions, and continuous audio streams. Extending this work to multi-label classification or sound event detection would move the field closer to practical applications. Finally, developing lightweight architectures optimized for edge devices could expand the accessibility of environmental sound recognition in smart homes, hearing assistive technologies, and embedded monitoring systems. Together, these directions represent meaningful avenues for advancing the reliability, scalability, and real-world impact of environmental sound classification systems.

References

- [1] Piczak, K. J. (2015). *ESC: Dataset for environmental sound classification*. In *Proceedings of the 23rd ACM International Conference on Multimedia* (pp. 1015–1018). <https://doi.org/10.1145/2733373.2806390>
- [2] Park, D. S., Chan, W., Zhang, Y., Chiu, C.-C., Zoph, B., Cubuk, E. D., & Le, Q. V. (2019). *SpecAugment: A simple data augmentation method for automatic speech recognition*. arXiv:1904.08779. <https://arxiv.org/abs/1904.08779>
- [3] Kong, Q., Cao, Y., Iqbal, T., Wang, Y., Plumbley, M. D., & Wang, W. (2020). *PANNs: Large-scale pretrained audio neural networks for audio pattern recognition*. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 28, 2880–2894. <https://arxiv.org/abs/1912.10211>
- [4] Gong, Y., Chung, Y.-A., & Glass, J. (2021). *AST: Audio spectrogram transformer*. arXiv:2104.01778. <https://arxiv.org/abs/2104.01778>

Student Name	Contributed Aspects	Details
Prashant Sah	Project Ideation, Research and Implementation of Baseline CNN	Originated the project idea and identified the ESC-50 dataset. Conducted initial literature review to define the problem scope and experimental direction. Designed and implemented the Baseline CNN architecture, built the full training and validation pipeline, and performed early experimentation and debugging to establish the foundation for all subsequent models.
Karan Shah	Researching and training ResNet18 and EfficientNet models	Background research on pre-trained image models. Training ResNet18 and EfficientNet, including validation, measurement of accuracy and comparison with baseline.
Jenny Li	Data-loading and transformation, time and pitch shifting, spectral augmentation.	Loaded and transformed 1D audio clips to 2D Mel-spectrograms, applied time and pitch-shifting, implemented spectral augmentation with best performing model (EfficientNet) along with training and validation post-augmentation.
Rahul Sati	Feature Ablation Study: design, implementation, and analysis.	Designed and ran an ablation study comparing raw waveform, mel-spectrogram, and MFCC inputs on ESC-50. Built a matched-capacity WaveformCNN and consistent torchaudio pipelines; trained all models under the same setup. Analyzed convergence, overfitting, and accuracy.

Table 1. Contributions of team members.